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About Me

Mathematics and Informatics graduate from the HUST Talent Program with two peer-reviewed international publications in graph algorithms and representation learning: WACV 2024 (CORE A) and RIVF 2025. My research focuses on random-walk algorithms, stochastic processes on networks, and graph machine learning for dynamic systems, alongside five years of experience building production ML systems in banking and computer vision.

Research Interests

Random walks on graphs · stochastic processes on networks · dynamic and large-scale graph algorithms · graph neural networks · representation learning · computer vision · probabilistic modelling · machine learning for decision systems

Education

Hanoi University of Science and Technology, Hanoi, Vietnam **Bachelor in Mathematics and Informatics (Talent Program)**, 2017–2022

- Talent Program context: K62 cohort of 21 students with a dedicated Mathematics–Informatics curriculum.
- GPA: 3.28/4.0 (8.2/10).
- Thesis: *Network Model and Risk Assessment for the COVID-19 Pandemic in Vietnam*. Grade: A. Bachelor of Science Research Project, 8 credits.
- Code: covid19-risk-evaluation
- Relevant courses: Numerical Analysis (A+), Optimization Methods (A), Data Structures and Algorithms (A), Computation Programming (A+), Stochastic Models and Applications (A), Programming Techniques (A+), Object-Oriented Programming (A+), Decision Support System (A+), System Analysis and Design (A).
- Additional coursework: Social Network Analysis (University of California, Davis); Deep Learning with PyTorch: Generative Adversarial Networks (Coursera); Neo4j Graph Academy series covering Fundamentals, Cypher, Graph Data Modeling, and Building Neo4j Applications with Python.

Research Experience

Graph Community Detection for Dynamic and Large-Scale Graphs Institute of Mathematics, Vietnam Academy of Science and Technology · Apr 2025–present (part-time) Supervisor: Assoc. Prof. Phan Thi Ha Duong

- Develop scalable community-detection methods for dynamic graphs with 50,000+ nodes, targeting runtime and partition quality.
- Implement and optimize DF-Louvain community detection, achieving a 15–30× runtime improvement with Numba.
- Implement the random-walk-based refinement code, contribute to a partial proof of the paper’s separation theorem, and extend the argument to localize modularity change within a cluster; published at RIVF 2025.
- Build a graph-community benchmarking toolkit and optimized overlapping-modularity routines used in the RIVF 2025 experimental evaluation.

Spam Detection over Telephony Networks using Graph Neural Networks Faculty of Applied Mathematics and Informatics, Hanoi University of Science and Technology · Nov 2024–present (part-time) Supervisor: Assoc. Prof. Dr. Ngoc C. Le

- Develop graph-based methods to detect spam and fraud calls over telephony call-history graphs.
- Design topology-aware behavioural features that capture malicious calling patterns from network structure.
- Implement and benchmark graph neural network architectures against a prepared benchmark dataset and baseline suite. Ongoing research; dataset and manuscript are in preparation.

Graph-Based COVID-19 Risk Assessment Model National Steering Committee for COVID-19 Prevention and Control · 2021

- Built a Markov-chain random-walk model over approximately 10,600 administrative units of Vietnam, using 30,000 Monte Carlo simulations to estimate regional outbreak risk under uncertain patient mobility.
- Worked within the National Steering Team for COVID-19 Prevention and Control.
- Led an eight-student team constructing and maintaining the regional adjacency graph from public geographic data and local administrative changes.
- Processed and validated 10,000+ patient records and generated ward-level risk maps for response work during Vietnam’s fourth COVID-19 wave (Apr–Jul 2021).
- Recognition documented by a service card and a Certificate of Merit from the Executive Committee of the Hanoi Youth Union.
- Code: covid19-risk-evaluation

Long-Term Person Re-Identification Remote collaboration with University of Houston researchers · 2023–2024

- Joined the project as its first member and contributed to the graph-based gait/shape representation and fusion network for long-term person re-identification.
- Co-designed the viewpoint-aware contrastive loss to reduce identity variation across viewpoints; the work was published at WACV 2024 (CORE A).
- Implemented and maintained the public codebase in my personal GitHub repository, including repository architecture, configuration/factory system, skeleton-graph components, and MEBOW orientation integration.
- Code: CVSL_LReID

Poisson Hidden Markov Model for Overdispersed Count Data School of Applied Mathematics and Informatics, HUST · 2020

- Developed a Poisson HMM with Baum–Welch/EM estimation, forward–backward algorithms, and Viterbi decoding for overdispersed NYC traffic-count data.
- Code: GPD_HMM_MHNN

Work experience

Senior Data Scientist, Vietnam Technological and Commercial Joint Stock Bank (Techcombank), Hanoi · Apr 2026–present

- Develop recommendation models for the bank-wide Hyper Personalization program, supporting individualized product and content recommendations across retail customer touchpoints.
- Build signature-duplication detection for copy-paste and template-replication forgeries as part of an AI document-fraud pipeline spanning transaction records and contracts.

Data Scientist, Vietnam Joint Stock Commercial Bank for Industry and Trade (VietinBank), Dong Da, Hanoi · Mar 2025–Apr 2026

- Designed a graph-based recommendation system over corporate transaction networks using centrality and network-structure signals, increasing conversion rate from 0.7% to 3.21%.
- Built and maintained MLflow model tracking/registry and GPU-enabled Kubernetes MLOps infrastructure supporting internal ML workflows.

Machine Learning Engineer, VMO Holdings Technology Joint Stock Company, Cau Giay, Hanoi · Apr 2023–Sep 2024

- Fine-tuned BERT, T5, and GPT variants for conversation-quality assessment.
- Adapted LLaVA-style models with CLIP-ViT and Swin vision backbones for multimodal tasks.
- Engineered automated retraining and drift-detection pipelines, reducing forecast MAPE from 0.24 to 0.14.

AI Engineer, Grooo International Joint Stock Company, Cau Giay, Hanoi · Apr 2021–Apr 2023

- Built an end-to-end surveillance identity-matching pipeline using RetinaFace and vector databases, reaching 98% accuracy in internal evaluation.
- Developed eKYC components including OCR, face recognition, anti-spoofing, and graph-based document parsing.
- Implemented GNN-based telephony fraud detection with active learning and continuous graph updates, reaching 96% accuracy in internal evaluation.

Publications

- V. D. Nguyen, K. Khaldi, **D. Nguyen**, P. Mantini, and S. Shah, “Contrastive Viewpoint-aware Shape Learning for Long-term Person Re-Identification,” *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024, pp. 1030–1038. Paper · Code
- D. H. Do, **D. Nguyen**, and T. H. D. Phan, “Improving the DF-Louvain Algorithm through Random Walk-Based Refinement,” *RIVF International Conference on Computing and Communication Technologies*, 2025, pp. 932–937. DOI · Code

Google Scholar: 49 citations, h-index 1 (accessed 2026-05-15).

Selected Projects

- Graph-Based Epidemic Model Simulation
- Network Model for COVID-19 Risk Evaluation in Vietnam
- Poisson Hidden Markov Models for Overdispersed Counts
- Contrastive Viewpoint-Aware Shape Learning for Long-Term Person Re-Identification
- Random-Walk Graph Partitioning – Dynamic Frontier

Teaching Experience

Teaching Assistant (part-time), PlusPlus Academy, Dong Da, Hanoi · Apr 2021–Oct 2021

- Taught Python data-science and machine-learning libraries, including NumPy, Pandas, Scikit-learn, PyTorch, and TensorFlow, to a class of 15 students.
- Guided students through algorithm implementation and evaluated final machine-learning and computer-vision projects.

Awards, Honors & Activities

- **Third Prize**, Vietnamese Mathematical Olympiad (VMO), 2017 — awarded by the Ministry of Education and Training.
- **Champion**, MLOps Marathon, 2023 — national MLOps competition organized by Open Factor Foundation; grand prize: 100,000,000 VND, among 121 opening-phase entrants.
 - Designed the serving architecture for the five-person team: separated API and model-worker processes, added Redis caching, distributed inference with RabbitMQ and Celery, and optimized JSON serialization/deserialization for 95th-percentile latency targets.
 - Scoring combined model accuracy (45%), system performance (45%), and drift detection (10%) across three progressive data-challenge phases.
- **Third Prize**, 38th Student Scientific Research Conference, School of Applied Mathematics and Informatics, HUST, 2021.
- **Certificate of Merit**, Executive Committee of the Hanoi Youth Union, Decision No. 2244 QĐ/TĐTN-VP, 30 September 2021, for achievements in COVID-19 prevention and control in Hanoi.
- **First Prize**, Mathematical Modeling Competition, Vietnam Mathematical Society, 2016.
- **Third Prize**, International Mathematics Tournament of the Towns, 2016.
- **Certificate of Distinction**, American Mathematics Contest 12 (AMC 12), AIME Qualifier, Mathematical Association of America, 2016.
- **Third Prize**, Province-level Mathematics Competition for High School Students, 2015.
- Sacombank Scholarship for academic excellence in mathematics, 2016.

Service and Leadership

- Technical Program Committee reviewer, RIVF 2025 — reviewed four papers.
- Invited to reviewer pools of CVPR, ECCV, and NeurIPS 2026.
- Head of Academic Committee, Hanoi Mathematical Modeling, 2020 — led a 10-member team collaborating with university faculty on problem design for high-school mathematical-modelling outreach.

Technical Skills

- **Programming:** Python, C++
- **Machine learning:** PyTorch, PyTorch Geometric, PyTorch Lightning, Scikit-learn, Optuna
- **Methods:** graph algorithms, graph neural networks, stochastic processes, Markov chains, Monte Carlo simulation, hidden Markov models, optimization, computer vision, representation learning
- **MLOps and infrastructure:** Docker, Kubernetes, MLflow, RabbitMQ, Kafka, Git, Metabase
- **Databases and graph tools:** Neo4j, Qdrant, MongoDB, PostgreSQL, SQL Server, Redis
- **Spoken languages:** Vietnamese (native), English (IELTS 6.5)